SECTION 6: Training and Fine-Tuning

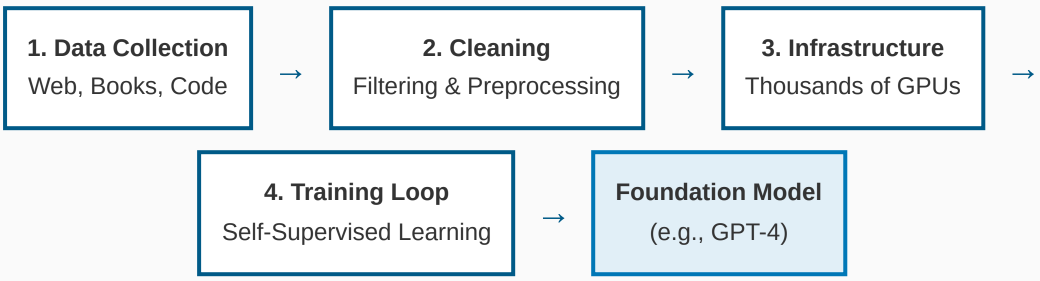
In this section, we pull back the curtain to understand how generative AI models are built. We explore the massive undertaking of pre-training, the customization power of fine-tuning, and the critical importance of data quality. Understanding these processes explains why models behave the way they do and how they can be adapted for specific needs.

Lecture 6.1: The Pre-Training Process

Building Foundation Models from Scratch

Pre-training is the initial, computationally intensive phase where a model learns general knowledge, language patterns, and reasoning capabilities from vast amounts of data. It creates a "foundation model" that serves as a base for many downstream applications.

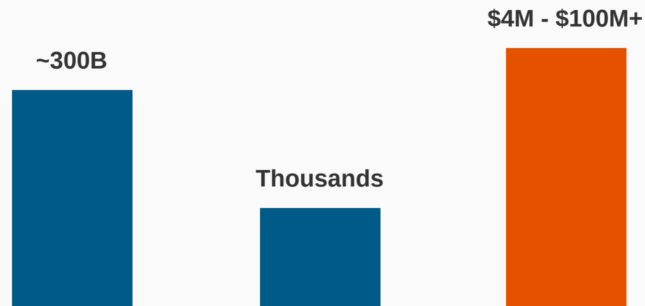
Figure 1: The Pre-Training Pipeline

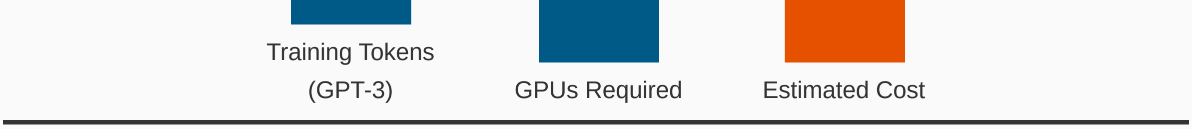


The Scale of Pre-Training

The resources required for pre-training are immense, following **Scaling Laws** which suggest that performance improves predictably with more data, parameters, and compute.

Figure 2: Resource Scale Comparison (Conceptual)





Why Pre-Training is So Expensive

1. Data Volume: Processing terabytes of text (e.g., Common Crawl, Wikipedia, GitHub).
2. Compute Power: Requires massive clusters of high-end GPUs running for weeks or months.
3. Energy: Consumes megawatts of electricity, raising sustainability concerns.
This is why mostly large tech companies (OpenAI, Google, Meta) build foundation models.

The Training Loop: Self-Supervised Learning

Models learn through **next-token prediction**. They don't need human labels for everything; the data itself provides the supervision.

- **Objective:** Predict the next word in a sentence based on context.
- **Result:** By learning to predict effectively, the model internalizes grammar, facts, reasoning patterns, and even some world knowledge.

LECTURE 6.1 KEY TAKEAWAY

Pre-training is a massive, one-time investment to create a general-purpose "smart" model. It learns broadly from the internet but inherits biases and errors from that data. This foundation enables all specific AI applications.

Lecture 6.2: Fine-Tuning Explained

Adapting the Foundation

Fine-tuning adapts a pre-trained model for a specific task or domain. It's like taking a generalist college graduate and giving them specialized job training.

Aspect	Pre-Training	Fine-Tuning
Goal	General knowledge & capabilities	Specific task performance
Data Size	Billions of examples	Thousands to hundreds of thousands

Cost/Time	Millions of $, Weeks/Months	Hundreds of $, Hours/Days
Outcome	Base Model (e.g., GPT-3)	Specialized Model (e.g., ChatGPT)

Types of Fine-Tuning

- **Full Fine-Tuning:** Updating all model parameters. Effective but resource-intensive and risks "catastrophic forgetting" (losing general knowledge).
- **Parameter-Efficient Fine-Tuning (PEFT):** Updating only a small subset of parameters (e.g., **LoRA** - Low-Rank Adaptation). Much faster and cheaper.

Specific Methods

1. **Instruction Tuning:** Teaching the model to follow commands (e.g., "Summarize this," "Translate this") rather than just continuing text.
2. **Domain Adaptation:** Specializing for medicine, law, or coding by training on industry-specific documents.
3. **RLHF (Reinforcement Learning from Human Feedback):** Aligning the model with human values (helpfulness, safety).

Figure 3: RLHF Alignment Process

1. Model Generation
Model produces multiple responses to a prompt.

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2. Human Ranking
Humans rate responses from best to worst.

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3. Reward Model
A separate AI learns to predict human preferences.

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4. Policy Update (PPO)
The main model is updated to maximize reward scores.

When to Fine-Tune vs. Prompt Engineering

Use Prompt Engineering when: You need quick results, the task is general, or you have limited data.

Use Fine-Tuning when: You need to teach new knowledge (e.g., proprietary company data), specific complex behaviors, or a unique style/tone that prompts can't consistently achieve.

LECTURE 6.2 KEY TAKEAWAY

Fine-tuning makes general models useful and safe. It creates specialized tools from general foundations. Techniques like RLHF are crucial for making chatbots helpful and aligned with human intent.

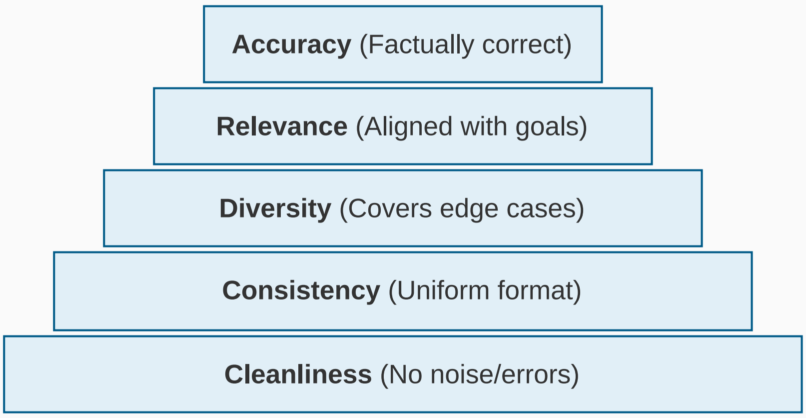
Lecture 6.3: Data Requirements and Quality

Data: The Fuel of AI

The capabilities, limitations, and biases of an AI model are fundamentally determined by its training data. The principle **"Garbage In, Garbage Out"** applies strictly.

Data Quality Dimensions

Figure 4: The Data Quality Hierarchy



The Problem of Bias

Data bias is a major challenge because models amplify existing societal biases found in training data.

- **Representation Bias:** Underrepresenting certain groups or geographies.

- **Stereotypical Bias:** Associating certain roles or traits with specific genders or ethnicities.
- **Historical Bias:** Reflecting past inequalities present in historical data.

Data Sources

Source	Pros	Cons
Web Scraping (Common Crawl)	Massive scale, diverse topics	High noise, misinformation, toxic content
Books & Papers	High quality, long-form, reasoning	Copyright issues, limited access
Code Repositories (GitHub)	Teaches logic & syntax	May contain bugs or bad practices
Synthetic Data	Unlimited, controlled, privacy-safe	Risk of "model collapse" if overused

⚠️ Legal, Privacy & Ethical Issues

- **Copyright:** Lawsuits from authors and artists (e.g., NYT vs OpenAI) regarding fair use of training data.
- **Privacy:** Scraping may inadvertently include PII (Personally Identifiable Information). Models must be scrubbed.
- **Consent:** Lack of opt-in mechanisms for creators whose work feeds the models.

Data Curation

Modern training involves extensive **filtering** to remove low-quality, toxic, or duplicative content. **Data Documentation** (like Datasheets for Datasets) is becoming standard to ensure transparency about data origins and limitations.

LECTURE 6.3 KEY TAKEAWAY

Data quality is often more important than model architecture. High-quality, diverse, and clean data produces capable and fair models. Legal and ethical sourcing of data is the next major frontier in AI development.